

Properties of determinant

Every square matrix have associated with a number called determinant.

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Properties:

I. The determinant of the identity matrix is 1.

$$\det I = 1, \quad \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1, \quad \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1, \text{ and ...}$$

II. The determinant changes sign when two rows are exchanged.

$$\det \begin{bmatrix} c & d \\ a & b \end{bmatrix} = \begin{vmatrix} c & d \\ a & b \end{vmatrix} = - \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$\det P = 1$ or -1 , depending on the number of row exchange(even/odd).

III. The determinant depends linearly on the row. Linear combinations involve two operations—adding vectors and multiplying by scalars. Therefore this rule can be split into two parts:

$$a. \text{ Add vectors in row 1: } \begin{vmatrix} a + \acute{a} & b + \acute{b} \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} \acute{a} & \acute{b} \\ c & d \end{vmatrix}$$

$$b. \text{ Multiply by } t \text{ in row 1: } \begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Properties IV–X will be deduced from the properties I-III.

IV. If two rows of A are equal, then $\det A = 0$.

Exchange those rows $\Rightarrow$ same matrix but its determinant sign change (Prop. II).

So, $2 \det A = 0 \Rightarrow \det A = 0$

Properties of determinant

V. Subtracting a multiple of one row from another row leaves the same determinant.

$$\begin{vmatrix} a & b \\ c - la & d - lb \end{vmatrix} \xrightarrow{\text{III.a}} \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a & b \\ -la & -lb \end{vmatrix} \xrightarrow{\text{III.b}} \begin{vmatrix} a & b \\ c & d \end{vmatrix} - l \begin{vmatrix} a & b \\ a & b \end{vmatrix} \xrightarrow{\text{IV}} \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

VI. If A has a row of zeros, then $\det A = 0$.

$$\begin{vmatrix} 0 & 0 \\ c & d \end{vmatrix} \xrightarrow{\text{III.b}} 0 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 0$$

VII. The determinant of a triangular matrix U is multiplication of its diagonal elements.

$$\begin{vmatrix} d_1 & a & \dots & b \\ 0 & d_2 & \dots & c \\ & & \dots & \\ 0 & 0 & \dots & d_n \end{vmatrix} \xrightarrow{\text{V}} \begin{vmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ & & \dots & \\ 0 & 0 & \dots & d_n \end{vmatrix} \xrightarrow{\text{III.b}} d_1 d_2 \dots d_n \begin{vmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ & & \dots & \\ 0 & 0 & \dots & 1 \end{vmatrix} \xrightarrow{\text{I}} d_1 d_2 \dots d_n$$

VIII. If A is singular, then $\det A = 0$. If A is invertible, then $\det A \neq 0$.

If A is singular, elimination leads to a zero row in U. Then $\det A = \det U = 0$. If A is nonsingular, elimination puts the pivots $d_1 d_2 \dots d_n$ on the main diagonal.

IX. $\det AB = \det A \times \det B$

$$\det A^{-1} = \frac{1}{\det A}; \quad \det A^2 = (\det A)^2; \quad \det 2A = 2^n \det A;$$

X. The transpose of A has the same determinant as A itself: $\det A^T = \det A$.

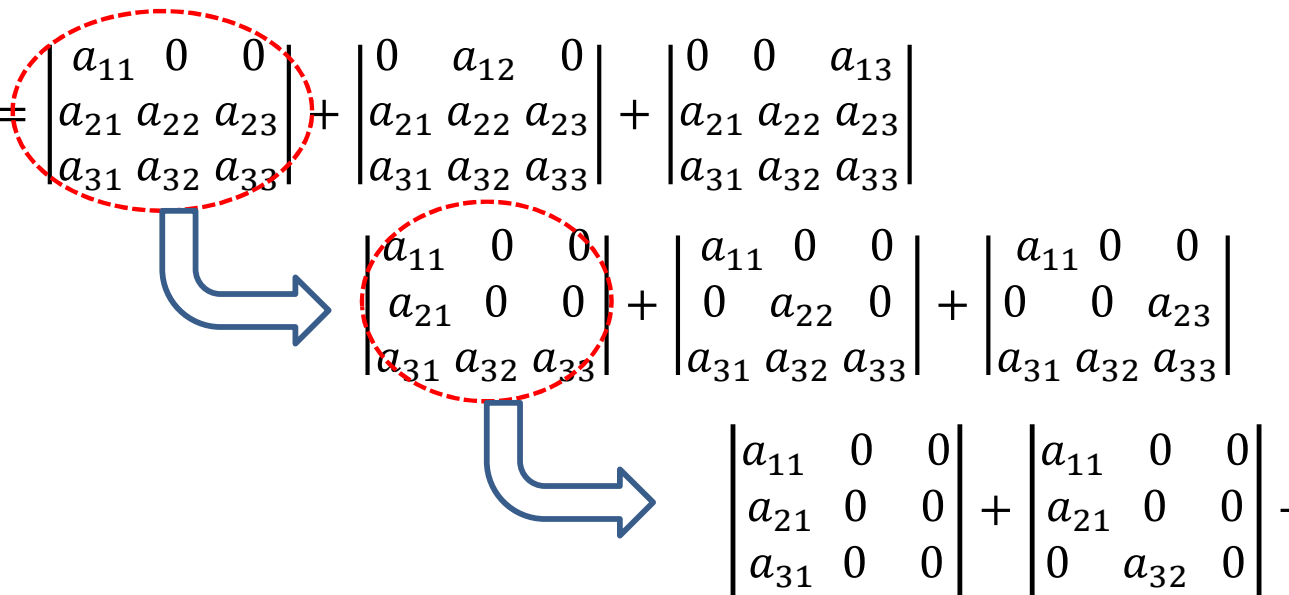
$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}, \text{ i.e., all properties are equally true for column.}$$

Determinant formula

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a+0 & b+0 \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c+0 & d+0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ c+0 & d+0 \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}$$

$$= \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} = \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} - \begin{vmatrix} c & 0 \\ 0 & b \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$



$$\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & 0 & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ a_{31} & 0 & 0 \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ 0 & 0 & a_{33} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & 0 & a_{23} \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ 0 & 0 & a_{23} \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ 0 & 0 & a_{23} \\ a_{31} & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & 0 \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & 0 \\ a_{31} & 0 & 0 \end{vmatrix}$$

$$= a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} + a_{31}a_{12}a_{23} + a_{21}a_{32}a_{13} - a_{31}a_{22}a_{13}$$

Big Formula:

$$\det A = \sum_{n!} \pm a_{1\alpha} a_{2\beta} a_{3\gamma} \cdots a_{n\delta}, \text{ where } (\alpha, \beta, \gamma, \dots, \delta) \text{ is permutation of } (1, 2, 3, \dots, n).$$

Cofactors

$$\begin{aligned}
 \det A &= a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} + a_{31}a_{12}a_{23} + a_{21}a_{32}a_{13} - a_{31}a_{22}a_{13} \\
 &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\
 &= \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & a_{23} \\ a_{31} & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & 0 \end{vmatrix}
 \end{aligned}$$

$i + j$ is even

$i + j$ is odd

Cofactor of $a_{ij} = \pm \det(n - 1 \text{ matrix with row } i \text{ and column } j \text{ errased}) = C_{ij}$.

Cofactor formula:

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

Example: Find the determinant of the matrix

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Inverse

Compute inverse of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix} = \frac{C^T}{\det A}$$

Check that $A \frac{C^T}{\det A} = I$ or $A C^T = (\det A)I$

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} C_{11} & \cdots & C_{n1} \\ \vdots & & \vdots \\ C_{1n} & \cdots & C_{nn} \end{bmatrix} = \begin{bmatrix} \det A & \cdots & 0 \\ \vdots & \det A & \vdots \\ 0 & \cdots & \det A \end{bmatrix} = (\det A)I$$

$$a_{11}C_{21} + a_{12}C_{22} + \cdots + a_{1n}C_{2n} = 0$$

Example: Find the inverse of $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} \text{cofactor } C_{11} & \text{cofactor } C_{21} \\ \text{cofactor } C_{12} & \text{cofactor } C_{22} \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

Cramer's rule

$$\text{Solve } Ax = b, \quad x = A^{-1}b = \frac{C^T b}{\det A} = \frac{\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ \vdots & \vdots & \cdots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}}{\det A}$$

Cramer's rule

Let $B_j = \begin{bmatrix} a_{11} & a_{12} & b_1 & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & b_n & a_{nn} \end{bmatrix}$ has b in column j . Now $\det B_j = b_1 C_{1j} + b_2 C_{2j} + \cdots + b_n C_{nj}$ which is equal to the j^{th} element of $C^T b$.

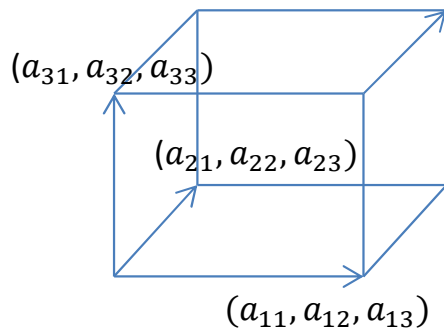
The j^{th} component of $x = A^{-1}b$ is the ratio $x_j = \frac{\det B_j}{\det A}$, where $B_j = \begin{bmatrix} a_{11} & a_{12} & b_1 & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & b_n & a_{nn} \end{bmatrix}$ has b in column j .

Example: Solve
$$\begin{array}{rcl} x_1 & + & 3x_2 = 0 \\ 2x_1 & + & 4x_2 = 6 \end{array}$$

According to the Cramer's rule $x_1 = \frac{\begin{vmatrix} 0 & 3 \\ 6 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}} = \frac{-18}{-2} = 9$ and $x_2 = \frac{\begin{vmatrix} 1 & 0 \\ 2 & 6 \end{vmatrix}}{\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}} = \frac{6}{-2} = -3$

Volume of a box

When all angles are right angles—the edges are perpendicular, and the box is rectangular. Then the volume is the product of the edge lengths: $\text{volume} = l_1 l_2 \dots l_n$.



When the edges of that box are the rows of A . With right angles, these rows are orthogonal and $A^T A$ is diagonal:

$$A^T A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} = \begin{bmatrix} l_1^2 & 0 & 0 \\ 0 & l_2^2 & 0 \\ 0 & 0 & l_3^2 \end{bmatrix} = l_1^2 l_2^2 l_3^2$$

$$\det(A^T A) = \det(A^T) \cdot \det(A) = (\det A)^2 = (l_1 l_2 l_3)^2$$

$$\det A = \pm l_1 l_2 l_3.$$

If the angles are not 90° , the volume is not the product of the lengths. In the Figure, the “volume” of a parallelogram equals the base ℓ times the height h . The vector $b - p$ of length h is the second row $b = (a_{21}, a_{22})$, minus its projection p onto the first row. The key point is this: By rule 5, $\det A$ is unchanged when a multiple of row 1 is subtracted from row 2. We can change the parallelogram to a rectangle, where it is already proved that volume = determinant.

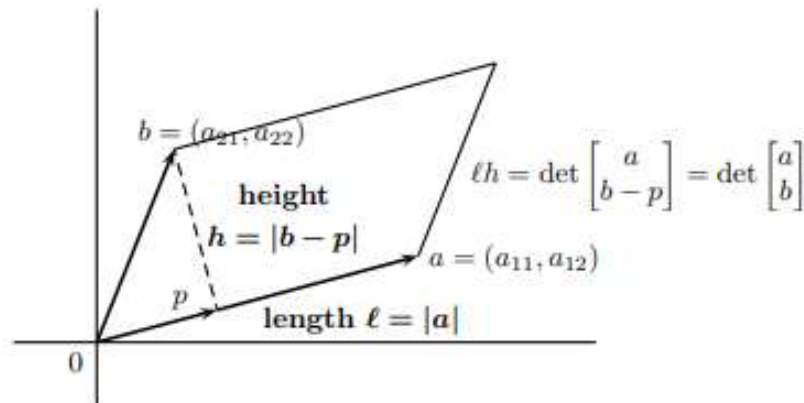


Figure : Volume (area) of the parallelogram = ℓ times $h = |\det A|$.